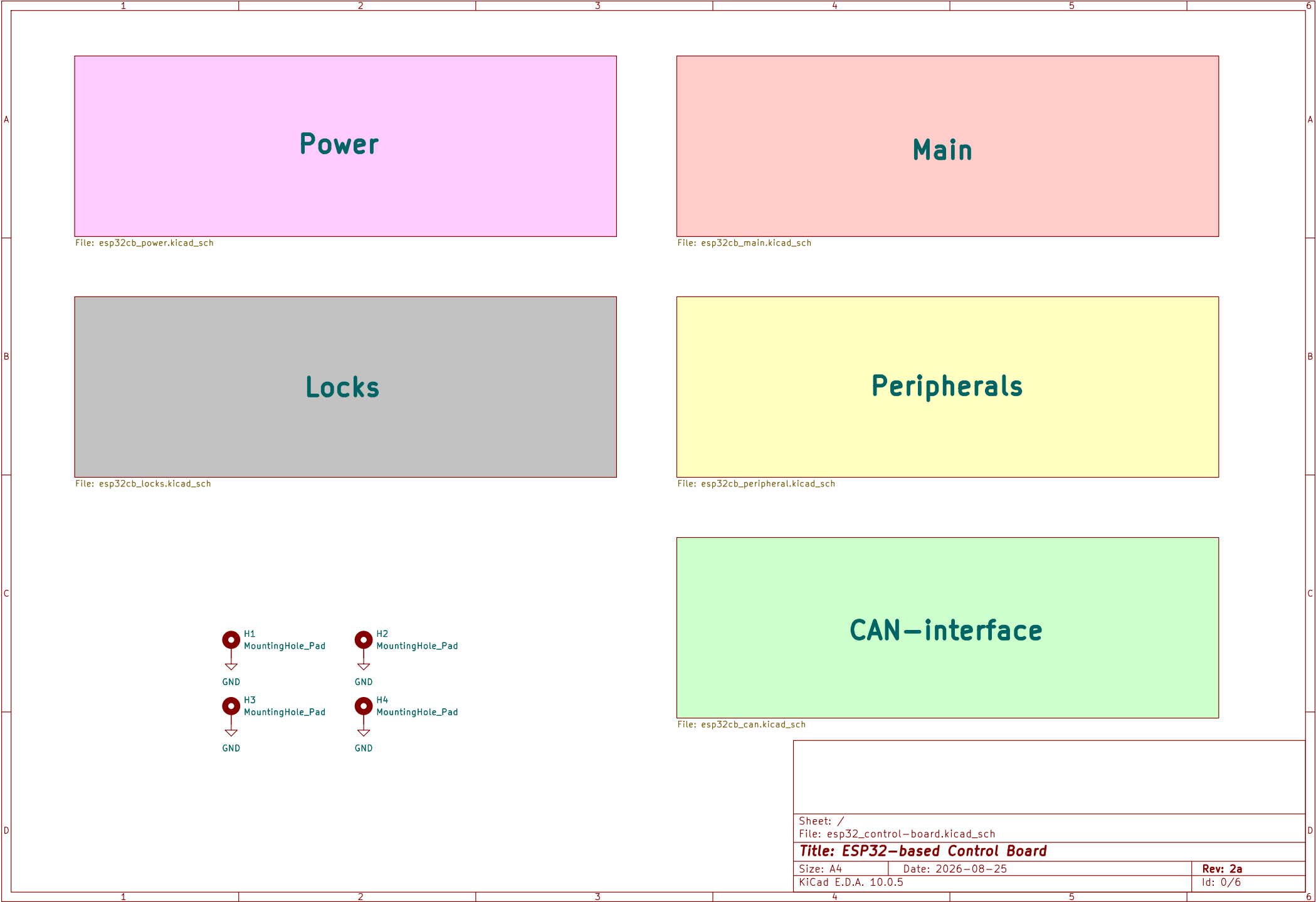
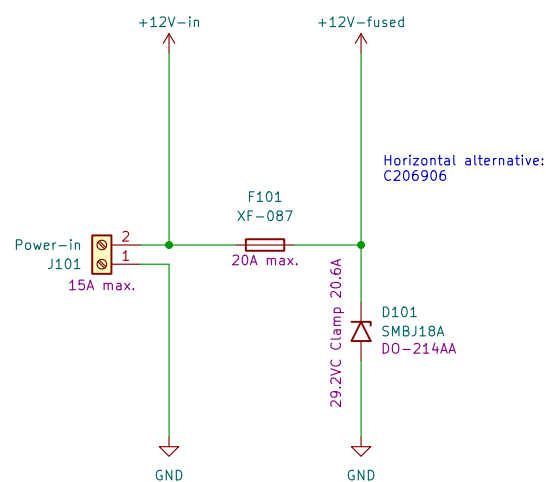
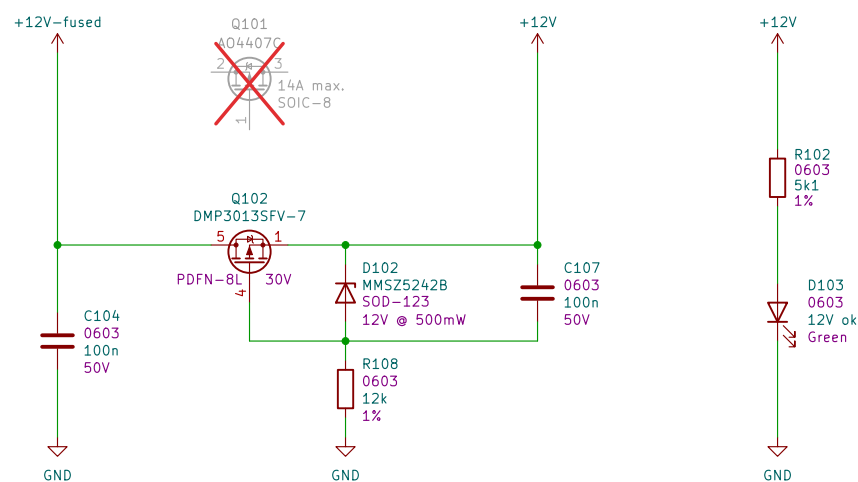


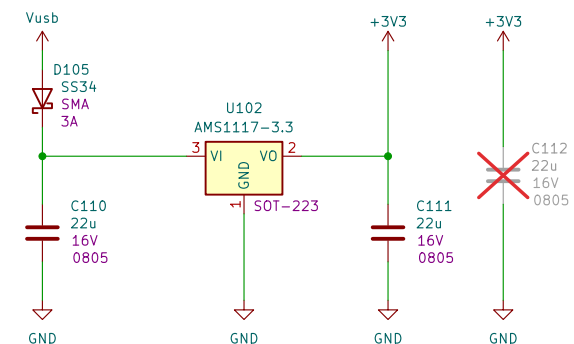




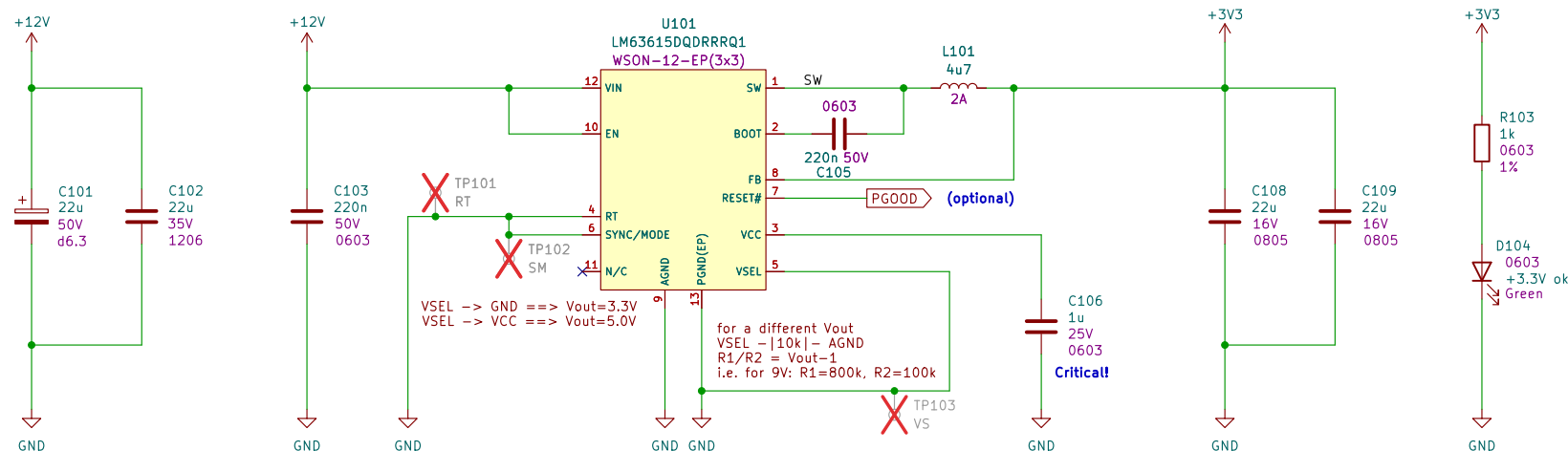
Overcurrent and overvoltage protection



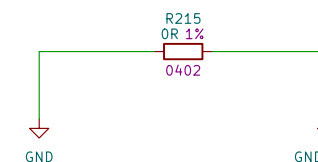
Reverse polarity protection



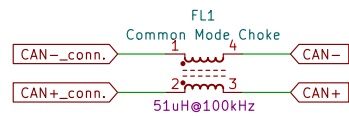
USB-line reverse-current protection



DC/DC buck converter



Power-GND and Digital GND connection

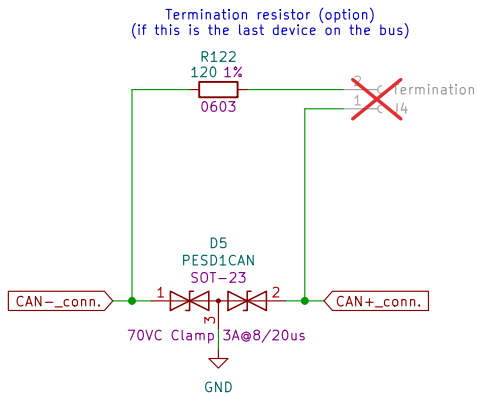


Alternative automotive CAN-chokes:

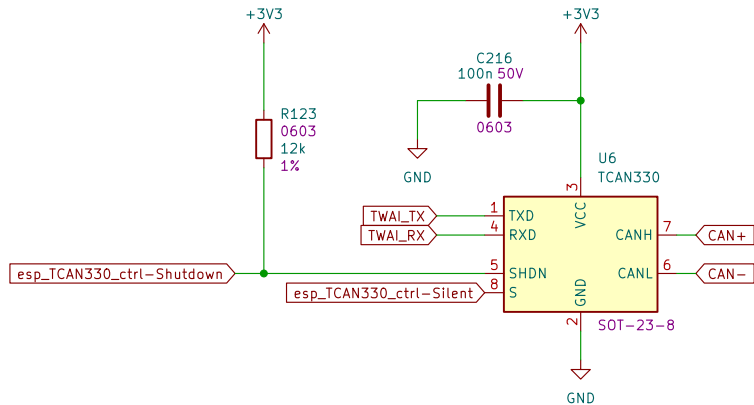
TDK ACT45B-510-2P ($\approx 51\Omega$ @100MHz)
Murata DLW43SH510XK2
Würth 744232101
Requirements:

2-line choke:
 $I_{min} \geq 100\text{ mA}$ ($I_{nom.} = 200\text{--}300\text{ mA}$);
 $R = 30\text{--}100\Omega$ @100 MHz

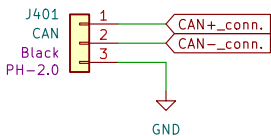
CAN-bus EMI-filtering



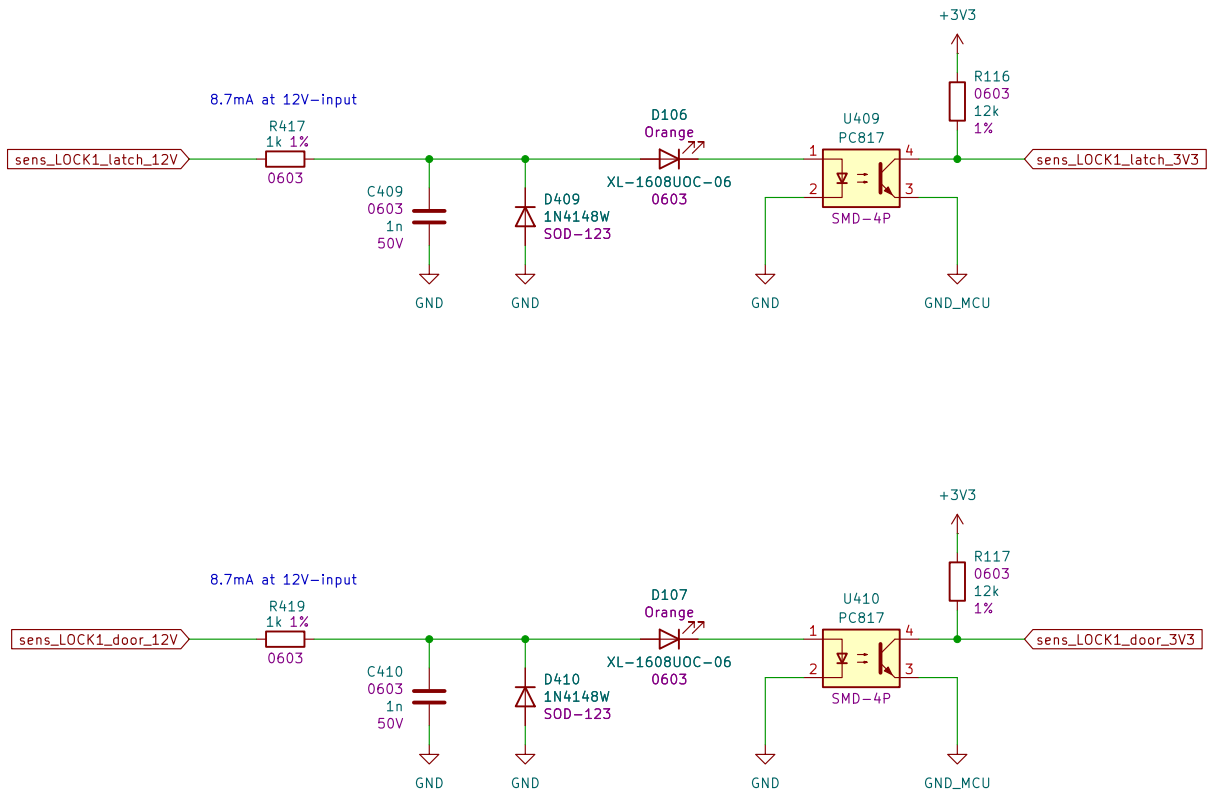
ESD-protection of CAN-line



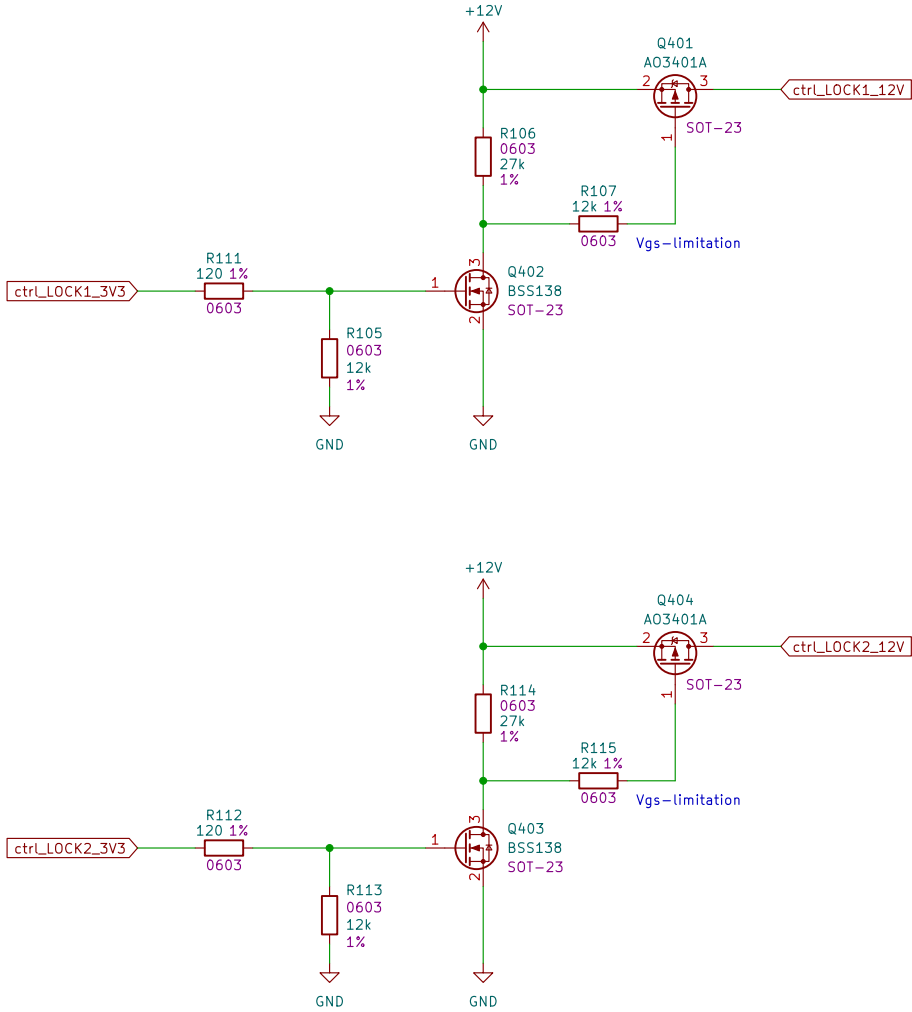
CAN-tranciever



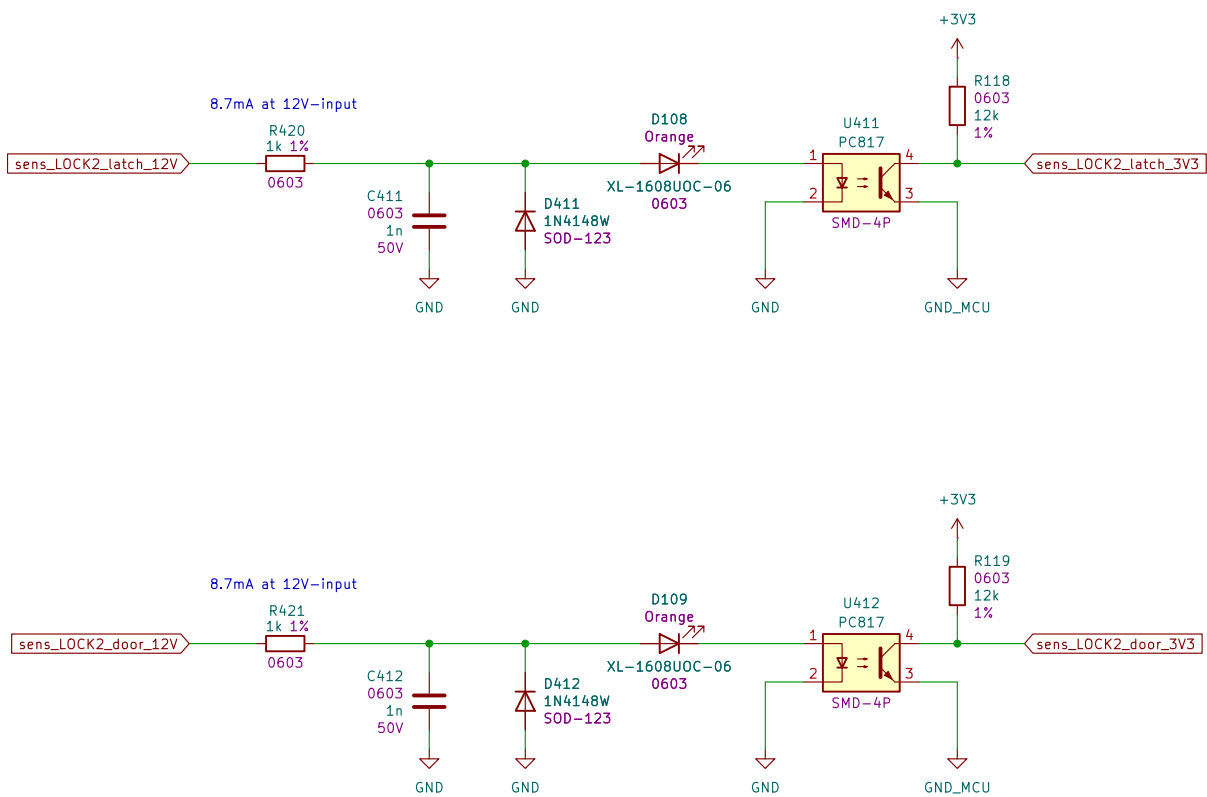
CAN-interface connector



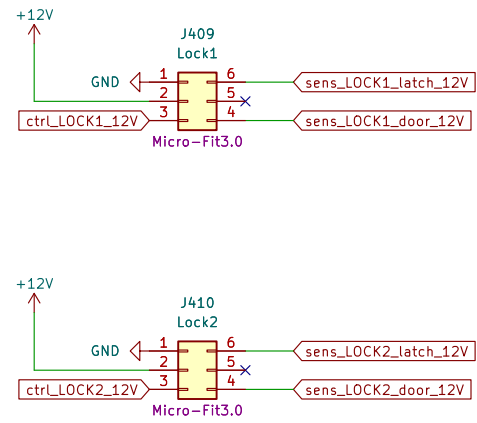
Lock1 sensors' processing circuits



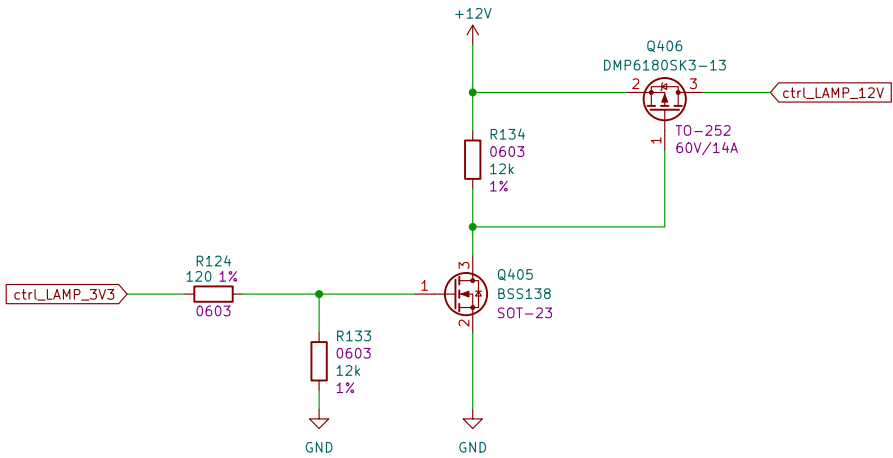
Locks' control circuits



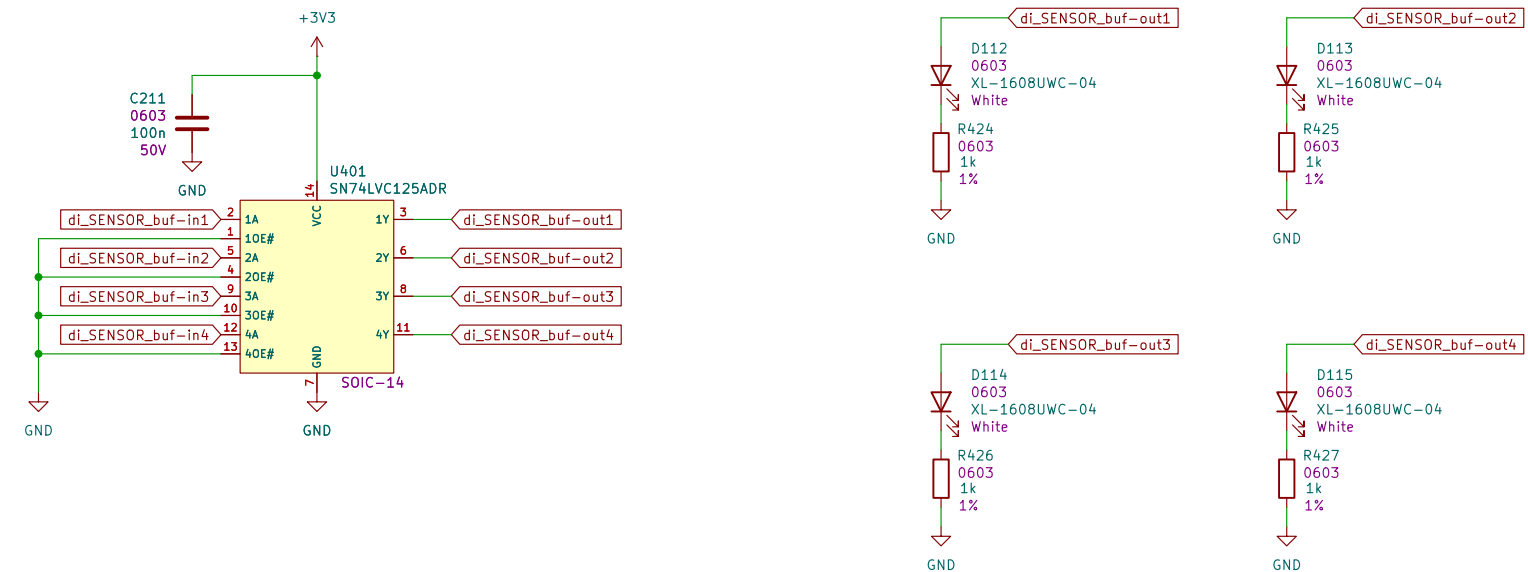
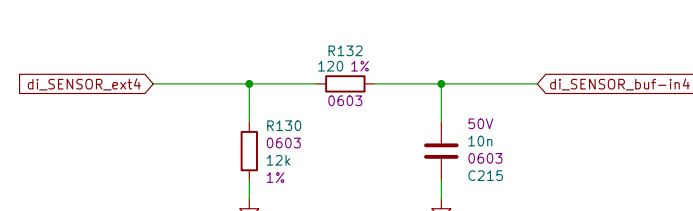
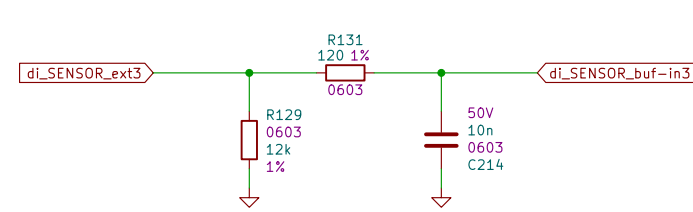
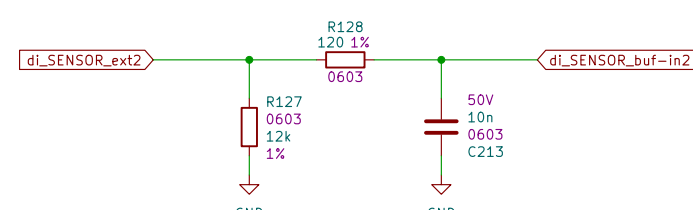
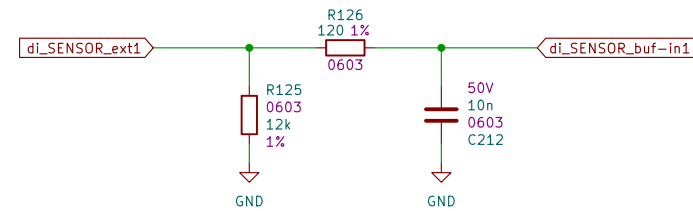
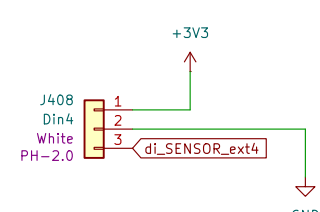
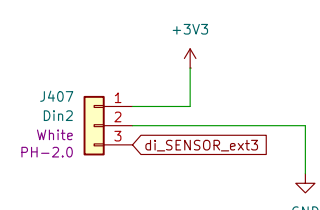
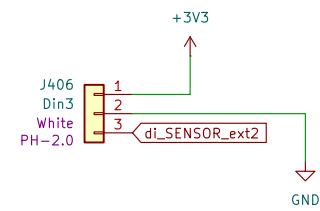
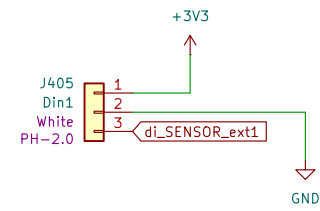
Lock2 sensors' processing circuits



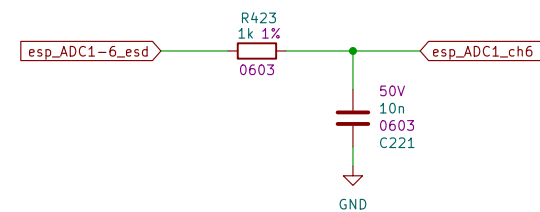
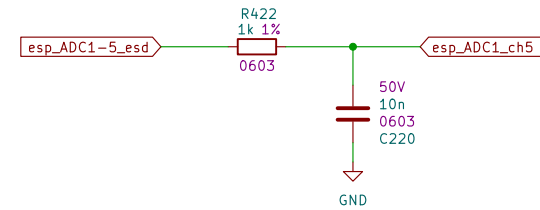
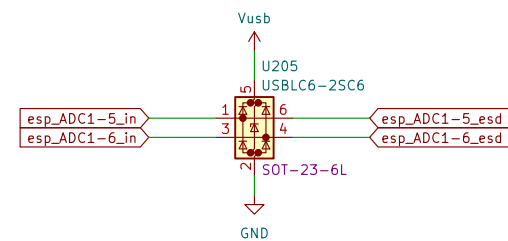
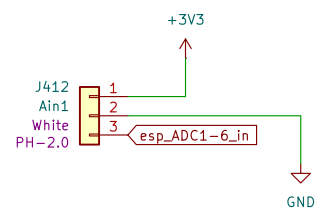
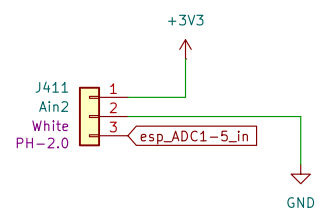
Locks' connectors



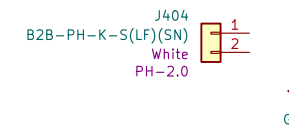
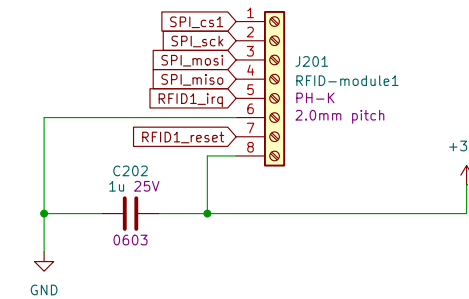
Lamp control circuit



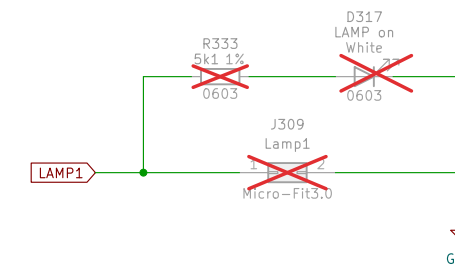
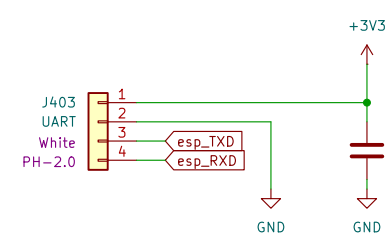
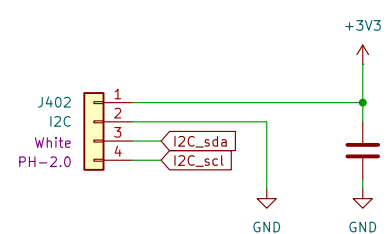
Digital Inputs



Analog Inputs



RFID-module connector



Interfaces